

Unified Risk Architecture

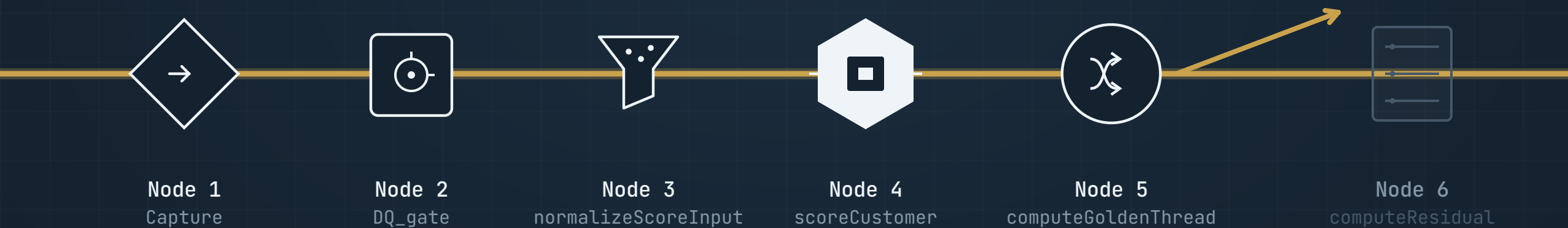
GLOBAL SCORING ENGINE • US PERIMETER : SYSTEM BLUEPRINT



AUTHORITATIVE MODEL : CRAM-US-2026-07-FREEZE-03

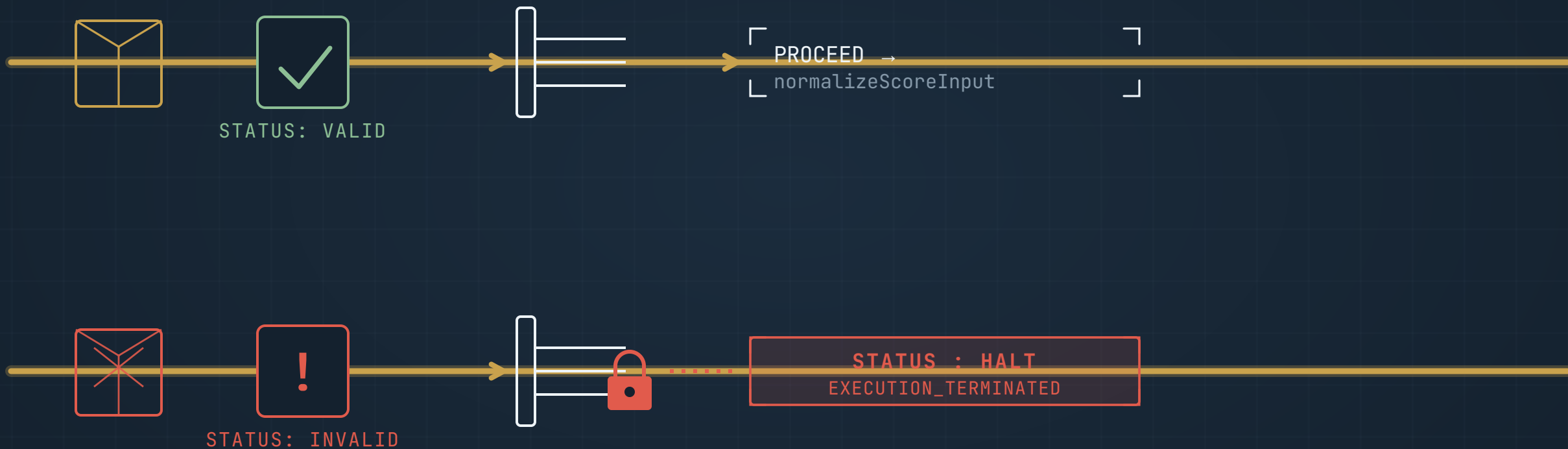
[CORE PIPELINE]

[MIS DISPLAY]

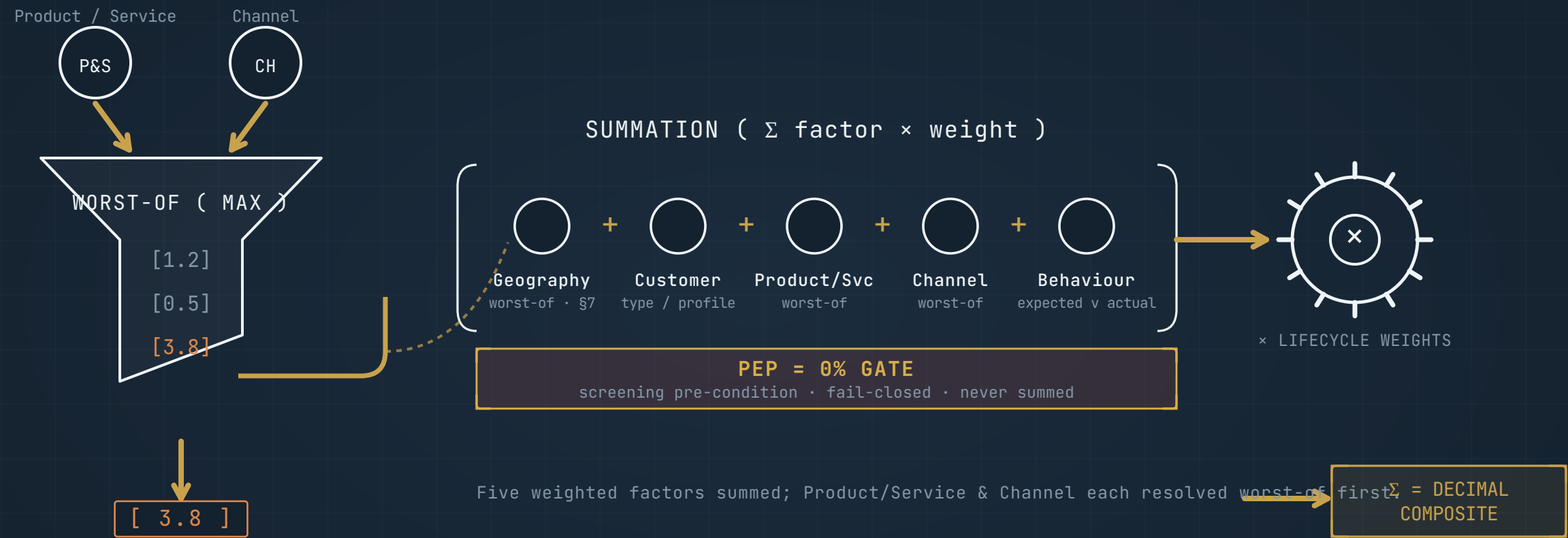


Capture → DQ_gate → normalizeScoreInput → scoreCustomer → computeGoldenThread → computeResidual

[NODE 01 → 02] : CAPTURE & DQ TOLLBOOTH



[NODE 04_A] : COMPOSITE EQUATION VISUALIZER



[NODE 04_B] : REGIONAL PARAMETER TOGGLE

UAE PERIMETER

Channel 25%

Geography 20%

US PERIMETER

Geography 25%

Channel 20%

UAE Geography – WEIGHTED AVERAGE

☐ Residence	25%	☐ SoW / SoF	15%
☐ Nationality	20%	☐ Corridor	15%
☐ Birth	10%	☐ Digital-Geo	15%

$\Sigma = 100\% \rightarrow$ weighted-average geography factor

Corridor / Digital-Geo = NULL
 $\rightarrow$ CONSERVATIVE DEFAULT Score 2.0 (UAE model only)

US Geography – WORST-OF (MAX) · residency-primary

No sub-weights – geography = single worst-of, not a weighted average.

✓ **Residence**

primary – residence / opco

✓ **Source of Wealth**

SoW

✓ **Source of Funds**

SoF

✓ **Incorporation**

entities only



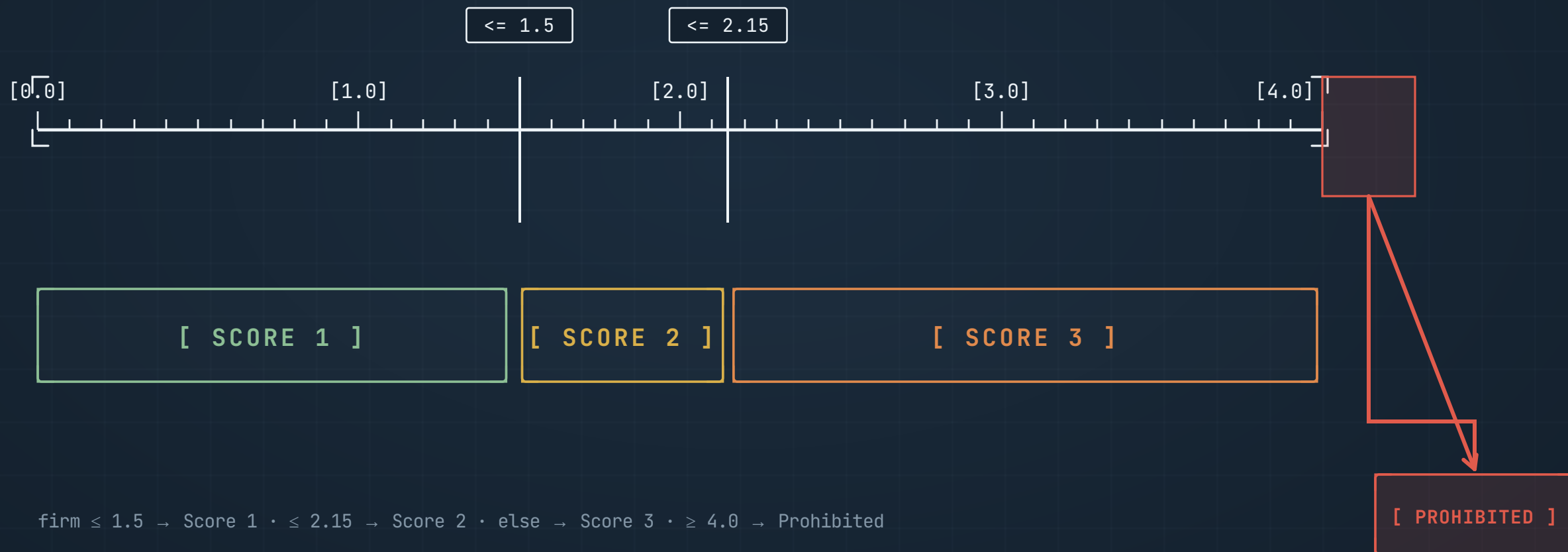
$\rightarrow$ firmToScore $\rightarrow$ clampScore

✗ Nationality – excluded (§11.1 non-discrimination)

✗ Country of birth – excluded (§11.1 · §7.4 Rule 3)

☐ Corridor tier – governance-only (§7.5) · Digital-Geo – not captured (§7.3)

[NODE 04_C] : firmToScore TRANSLATION



[NODE 04_D] : OVERRIDE COLLISION ENGINE

PRECEDENCE : PROHIBITED (override action) > HIGH > MEDIUM > LOW = math band

Three rating tiers only – Low / Medium / High. Prohibited is an action, not a tier. Moderate-High is treated as High. (CRAM §2.2 / §3.1)

PROHIBITED OVERRIDE ACTION not a tier Reject / Block / Exit	OVR-001 · Confirmed sanctions / TFS true match (SDN / UN / UAE Local / NACTA) OVR-002 · Prohibited jurisdiction nexus (residence / SoF / SoW / corridor) + Zenus prohibited-restricted country + prohibited business ↳ DPMS: knock-out ONLY IF on Zenus prohibited list (A3.5); else DPMS = profession-risk 4 (§5.1), not a knock-out also (§12 P1): shell bank · domestic correspondent · TPPP · unregistered MSB · inability to identify UBO · no lawful economic purpose basis: CRAM §12 Priority 1 · §7.1-7.4 · §6.4 · OFAC strict liability · 31 CFR 501.603/604 · Program §10 / A3.5						
HIGH mandated floors incl. Moderate-High ≤12-mo · 6 if MLRO	CONFIRMED SUSPICION / TF → High + MLRO · 6-mo review · SAR fired (CRAM §12 Priority 2) OVR-010 · Confirmed suspicion / existing STR-SAR → +MLRO · 6-mo review OVR-003 · Sanctioned-wallet / mixer exposure – any-% direct or terror/sanctions-tagged, mixer-routed (≥10% within ≤5 hops) ↳ direct SDN / terrorism true-match → Prohibited via OVR-001 HIGH FLOOR – equal rank (CRAM §12 Priority 3) <table border="0"> <tr> <td>OVR-008 · Foreign / IO / material-associate PEP (FATF R.12)</td> <td>OVR-011 · Score-3 country / High-corridor concentration</td> </tr> <tr> <td>OVR-009 · Confirmed adverse media (financial crime)</td> <td>OVR-012 · High-risk NOB</td> </tr> <tr> <td>OVR-004 · UBO concealment / high-risk UBO (31 CFR 1010.230)</td> <td>OVR-020 · Behaviour – device-share; 3+ investigations; pass-through</td> </tr> </table> also: opaque/nominee · stablecoin high-risk corridor · non-OFAC restricted-country SoW/SoF (§7.4) · permitted FI end-user pending EDD OVR-011 keeps Score-3 country as a floor by risk-appetite (stricter than FATF R.19 Tier-4/5 min; §5.2 25%-weighted input) · basis: FATF R.12 / R.20	OVR-008 · Foreign / IO / material-associate PEP (FATF R.12)	OVR-011 · Score-3 country / High-corridor concentration	OVR-009 · Confirmed adverse media (financial crime)	OVR-012 · High-risk NOB	OVR-004 · UBO concealment / high-risk UBO (31 CFR 1010.230)	OVR-020 · Behaviour – device-share; 3+ investigations; pass-through
OVR-008 · Foreign / IO / material-associate PEP (FATF R.12)	OVR-011 · Score-3 country / High-corridor concentration						
OVR-009 · Confirmed adverse media (financial crime)	OVR-012 · High-risk NOB						
OVR-004 · UBO concealment / high-risk UBO (31 CFR 1010.230)	OVR-020 · Behaviour – device-share; 3+ investigations; pass-through						
MEDIUM	OVR-016 · Domestic OR IO PEP – floor only; EDD WHERE the relationship is higher-risk (FATF R.12; FFIEC PEP) also (§12 P4): non-face-to-face residual · new high-risk product · moderate adverse · new entity high activity · NB: reconcile Program glossary flat rule						
LOW	Low / Medium / High taken from the weighted score where no floor fires · raw bands (§3.1): ≤1.5000 Low · ≤2.1500 Medium · >2.1500 High						

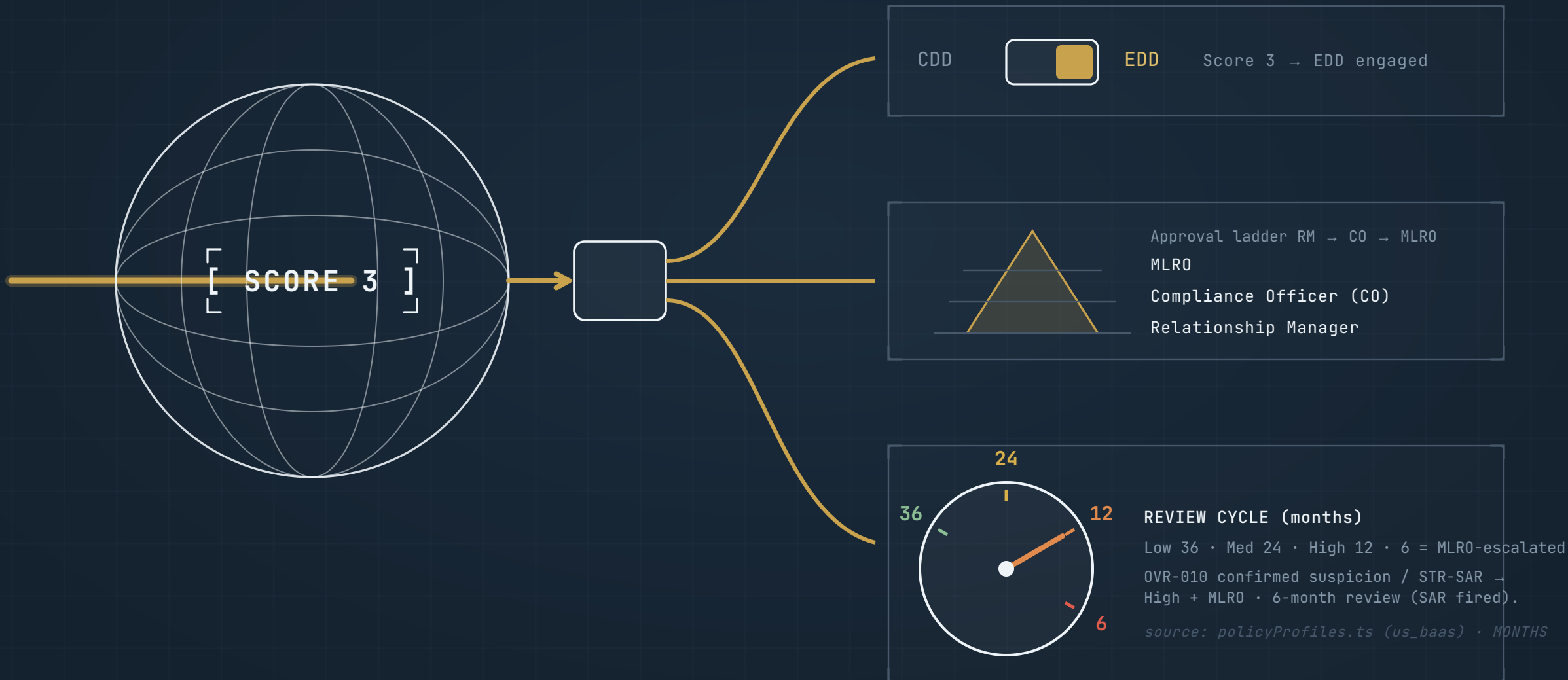
FINAL_RATING = MAX_FLOOR (MATH_BAND , ANY_TRIGGERED_OVERRIDE)

highest triggered floor wins – math band applies only when no floor fires; Prohibited overrides all

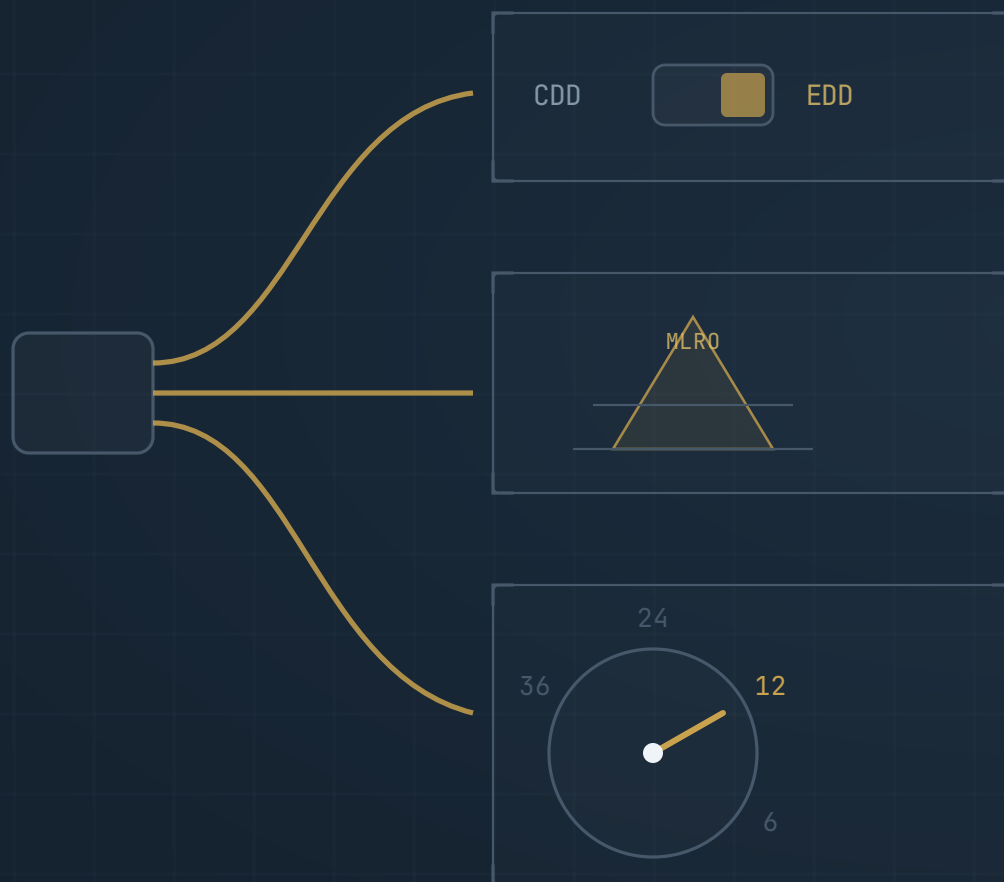
RCA (relative / associate) inherits the principal PEP's tier – not a separate floor (FATF R.12).

US perimeter: OVR-002 / OVR-011 read geoNexusFirmMax = MAX(residence, SoW, SoF, incorp?) – excludes nationality & birth.

[NODE 05_5] : GOLDEN THREAD ACTUATION

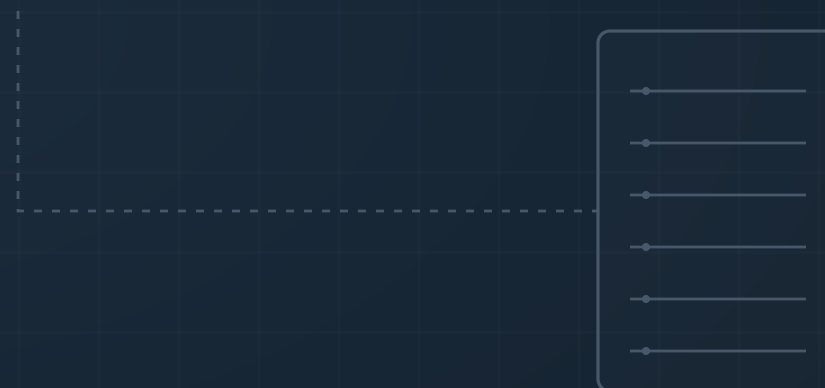


[NODE 06_] : COMPUTE RESIDUAL



// FOR MIS DISPLAY ONLY.
DOES NOT DRIVE INHERENT ACTUATION.

[RESIDUAL_RISK_COMPUTATION]



PART D : OPEN ARCHITECTURE & MODULE STATUS

DELIVERED

✓

Corridor Register

9-corridor typology board feeding TM cards #13 / #14 (Global Account)

status : SHIPPED

✓

UI Capture Fields

corridor capture surface live for Global Account onboarding

status : SHIPPED

PENDING

Corridor-tier scoring → composite

deliberately governance-only for now (\$7.5)

UAE 8-Pillar Country Library

country-risk library build outstanding

Digital-geolocation capture

IP / GPS / device geo not yet captured (\$7.3)

Scotermaps

exposure mapping module

Sub-Tier Approval

Compliance-Owner sign-off layer

SYSTEM_BLUEPRINT_UPDATED

v2.3 · aligned to CRAM-US-2026-07-FREEZE-03 · US perimeter (Zenrus rails)

WAITING_ON :

corridor-tier scoring · UAE library · digital-geo